Mikhail Razakov

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Quantitative Research | C++ · Python · Algorithms

Strong computer-science, algorithms, and research problem-solving background with hands-on C++ and Python, LLVM, and compiler/runtime engineering. I work in a hypothesis → experiment → validation loop using exact oracles, randomized/metamorphic tests, reproducible harnesses and benchmarks, counterexample analysis, and performance investigation. I work with JIT compilation, vectorization, x86-64 assembly, and generated/disassembled machine-code analysis.

Algorithms & competitions
HSE Vysshaya Proba prize-winner; overall winner of the MEPhI Junior competition in 2025 and 2026; Grand Prize at the Baltic science and engineering competition

Hypothesis → experiment
Exact oracles, randomized/metamorphic testing, adversarial counterexamples, reproducible harnesses, and benchmarks

C++ / low-level performance
C++23, LLVM, JIT, vectorization, x86-64 assembly, generated/disassembled machine-code analysis, and benchmarks

EXPERIENCE

July 1 — August 31, 2026

MCST — compiler engineering intern
- Implemented an LLVM loop-invariant code-motion pass in C++; reasoned about loop invariance, side effects, memory access, speculative safety, and loop structure.
- Built a Python verification harness comparing execution behavior before and after optimization and checking expected transformation / no-transformation cases.

2026

PS-form Memory Dependence Analyzer
- Ranked 1st of 49: the only 5.0/5.0 result and 104/104 official tests.
- Built an exact oracle for small domains, metamorphic/randomized checks, and reproducible wrong-answer/time-limit counterexamples.

2024 — present

UniversalToolchain / Wist2 — creator and primary developer
- Independently designed and developed a complex compiler/language system with intermediate representations, SSA/optimizations, and interpreter/CIL backends.
- Use benchmarking, JIT/CIL execution, and generated/disassembled-code analysis to investigate implementation behavior and performance.

PROJECTS

PS-form Memory Dependence Analyzer
Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.

UniversalToolchain / Wist2
The current exact test manifest passes with zero failures; interpreter/CIL parity, clean-consumer scenarios, and reproducible compiler/runtime contracts are verified.

x86-64 Codegen & Register Allocation Lab
A measurable pipeline with spill/reload metrics, code size, and reproducible result comparison.

SKILLS

Programming
C++23, Python, C17, C#, Rust

Algorithms & CS
data structures, graph algorithms, program analysis, CFG/SSA, compiler optimizations

Low-level & performance
LLVM, JIT/CIL, x86-64 assembly, vectorization, compiler optimizations, generated/disassembled machine-code analysis, benchmarking

Experiments & tooling
hypothesis formulation and validation, exact oracles, differential/metamorphic/randomized testing, Python harnesses, Linux, Git, CMake

EDUCATION

Software Engineering student at HSE University

RECOGNITION

HSE Vysshaya Proba prize-winner; overall winner of the MEPhI Junior competition in 2025 and 2026; first-degree diploma and Grand Prize at the Baltic science and engineering competition.

Moscow / remote